Introduction to Statistics

Lecture 1 - LAS

Don van den Bergh

2026-08-24

Today

- Introduction
- Course Outline
- Dictionary of Basic Concepts
- Role of Data in Scientific Research
- Measurement Level
- Descriptive Statistics
- Bivariate Descriptives

Introduction

Who are we?

Don van den Bergh

Amirali Rezazadeh

Why statistics?

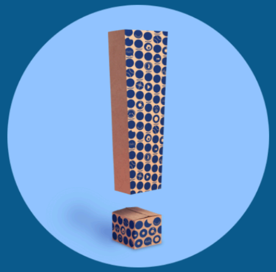
You grew up in the “digital age”. There is data about your entire life…

And everybody wants your data

bol.

Hallo Don

We informeren je over een beveiligingsincident bij CEVA Logistics, een logistieke partner waarmee bol samenwerkt voor de verwerking en bezorging van onze bestellingen.



Meer informatie

In het kort

Op 1 augustus heeft CEVA ons laten weten dat er een cyberaanval is geweest op hun systemen. Uit het onderzoek blijkt dat onbevoegden mogelijk toegang hebben gehad tot gegevens die worden verwerkt voor bestellingen vanuit één van de distributiecentra van bol.
Je ontvangt deze e-mail omdat jouw gegevens mogelijk onderdeel zijn van de informatie die tijdens deze cyberaanval is bekeken of gekopieerd.

What can you do with data?

- Data literacy is increasingly important in many jobs
- You can get insights from data to
  - Better understand social life
  - Predict sales and optimize marketing
  - Explore what activity in the brain is associated with observed behavior
- Data analysis is one of the most marketable concrete skills we teach at university

Course Outline

Overview

- What can you expect for this course?
  - 14 lectures
  - 10 lab sessions
- What will the exam look like?
  - Multiple choice exam (1-10)
  - SPSS test (Pass/ Fail)
- What more do we offer you?
  - Additional video lectures
  - Short lecture quizzes
  - Midterm quiz
  - Practice exams online
  - Practice workbook online

Learning Goals

- Compute and interpret commonly used descriptive statistics
- Recognize different probability distributions
- Explain the essential aspects of null-hypothesis significance testing
- Conduct statistical tests
- Choose the appropriate analysis technique for answering a specific research problem
- Use SPSS
- Draw valid conclusions from the results of empirical data analyses given specific research questions envisaged

Grading

- Two Exams (each 30%): on 15-10-2026 and 10-12-2026
- Two Assignments (Portfolio; 40%): due 13-10-2026 and 14-12-2026
- Combined resit for both exams (60%): 06-01-2027

Typical Schedule

- Monday: Lecture (either 10:45 - 12:30 or 16:45 - 18:30)
- Thursday/ Friday: Tutorial (12:45 - 14:30 or 14:45 - 16:30)
  - You can work on the weekly quizzes and the assignments

> **Note**
>
> Always check TimeEdit for the most up to date information!

GitBook

See https://cjvanlissa.github.io/stats12/

- Self study questions
- Weekly quizzes
- Assignments

GitBook – outdated!

All information under course-schedule is outdated.

- Classical lectures
- Dates are all from past year

Dictionary of Basic Concepts

Is Life Worth Living?

9 out of 10 adults in the Netherlands think life is worth living

Source: https://www.cbs.nl/nl-nl/nieuws/2026/31/9-op-de-10-volwassenen-vinden-het-leven-de-moeite-waard

Methods and Statistics

Methods is about the procedures of research:

- What actions do we take to collect good data
- Which participants to include
- How to measure what we want to measure
- How to design studies that are suitable for answering our research questions

Methods involve the design of a study or data collection

Methods and Statistics

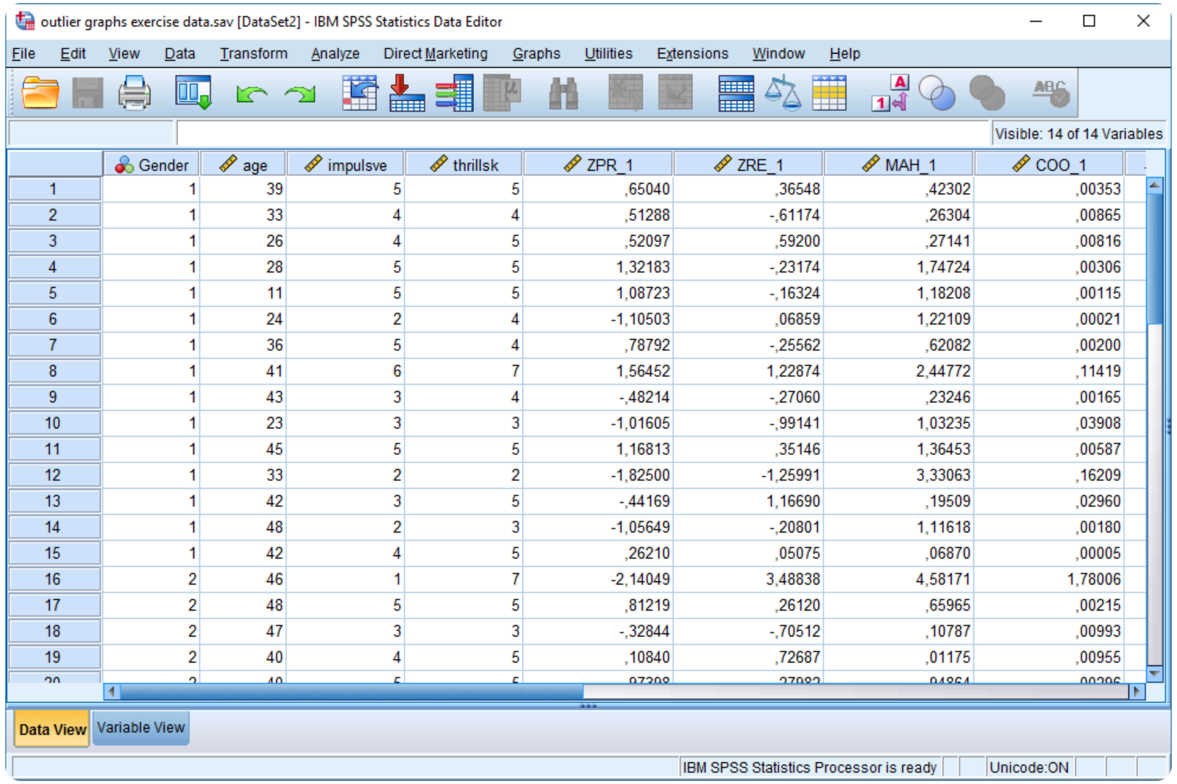
Statistics is about analyzing data

- **Descriptive statistics**: describing/summarizing data
- **Inferential statistics**: Making best guesses about the population from a smaller sample
- **Statistical modeling**: Representing a theory mathematically
- Predicting important outcomes (sales, well-being, neurological disorders)
- Exploring data to find interesting patterns
- Performing tests to answer theoretical research questions

Statistics are the mathematics behind going from *raw data* to conclusions.

Tabular Data

Most data in the social sciences is tabular, where each row represents an individual i observation, and each column j represents that individual's data on several variables:



	Gender	age	impulsve	thrillsk	ZPR_1	ZRE_1	MAH_1	COO_1
1	1	39	5	5	,65040	,36548	,42302	,00353
2	1	33	4	4	,51288	-,61174	,26304	,00865
3	1	26	4	5	,52097	,59200	,27141	,00816
4	1	28	5	5	1,32183	-,23174	1,74724	,00306
5	1	11	5	5	1,08723	-,16324	1,18208	,00115
6	1	24	2	4	-1,10503	,06859	1,22109	,00021
7	1	36	5	4	,78792	-,25562	,62082	,00200
8	1	41	6	7	1,56452	1,22874	2,44772	,11419
9	1	43	3	4	-,48214	-,27060	,23246	,00165
10	1	23	3	3	-1,01605	-,99141	1,03235	,03908
11	1	45	5	5	1,16813	,35146	1,36453	,00587
12	1	33	2	2	-1,82500	-1,25991	3,33063	,16209
13	1	42	3	5	-,44169	1,16690	,19509	,02960
14	1	48	2	3	-1,05649	-,20801	1,11618	,00180
15	1	42	4	5	,26210	,05075	,06870	,00005
16	2	46	1	7	-2,14049	3,48838	4,58171	1,78006
17	2	48	5	5	,81219	,26120	,65965	,00215
18	2	47	3	3	-,32844	-,70512	,10787	,00993
19	2	40	4	5	,10840	,72687	,01175	,00955

Example: Items

- I find my life worthwhile.
- I feel that I contribute something to society.
- I feel useful.
- I am able to do the things that I want to do and consider important in my life.
- How important are social contacts and relationships in your life?
- How important is personal development to you?
- How hopeful are you about the future? (1 - 10)

Example: Tabular Data

ID	AGE	AGE_GROUP	VOLUNTEER	WEEKLY_SOCIAL_CONTACT	WORTHWI
1	27	25 - 35	false	true	Strongly Ag
2	70	65 - 75	false	true	Strongly Ag
3	30	25 - 35	true	true	Strongly Ag
4	42	35 - 45	false	true	Strongly Ag

simulated based on the article

About Observations: Population

- The complete set of objects of interest
  - E.g., all people in NL, all students in this class
- Number of individuals in the population: N
  - N = 17.53 million, N = 75
- The correlation (linear dependence) between language and mathematics ability across all humans on earth
  - $\rho = .3$
- The “true” effect of class attendance on grades in students
  - $\delta = 0.4$

Population: Example

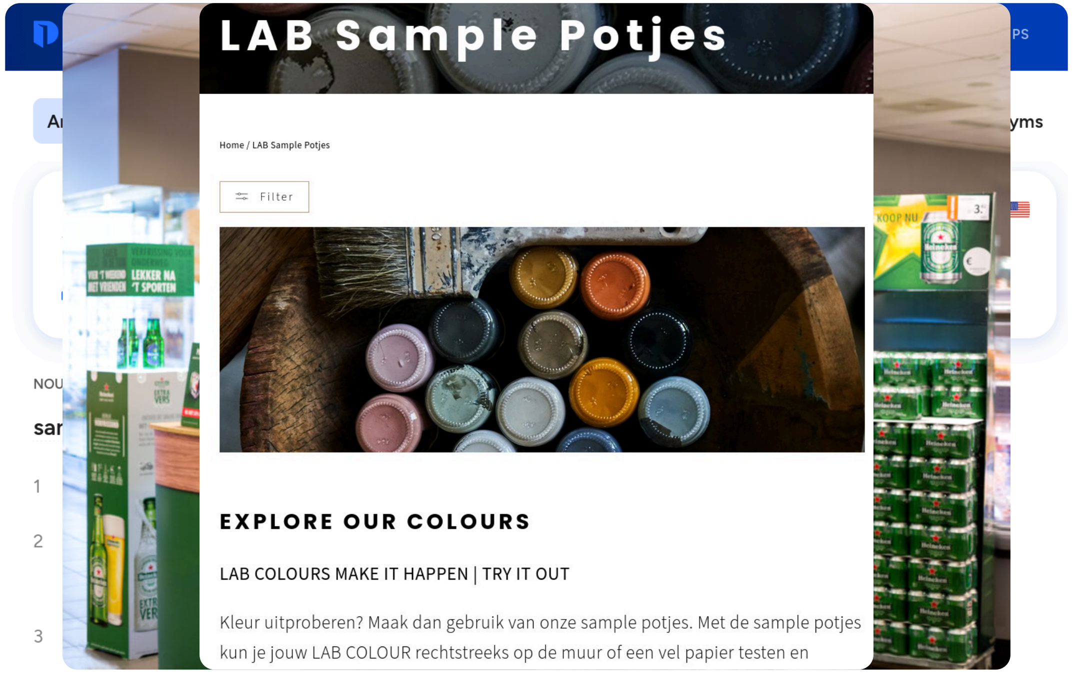
In the survey on happiness, the CBS interviewed 7,551 participants aged 15 and above.

What is the population?

Adults in the Netherlands / Above 15 year olds.

About Observations: Sample

What is a sample?



About Observations: Sample

- Observed part of the population
- Number of observations: *n*
- The correlation (linear dependence) between language and mathematics ability across all pupils measured as part of a study:
  - $r = .25$
- The effect size of class attendance on grades observed at Tilburg University
  - $d = 0.4$

A common pattern is *upper case* or *Greek* symbols (σ) for the population and *lower case* or *Roman* symbols (s) for the sample.

About Variables: Construct

Construct:

- Abstract feature of interest for the population
  - E.g., Short Term Memory, intelligence, perseverance, education

Operational definition:

- Concrete measurable representation of the construct
  - E.g., Number of words recalled, the Wechsler Adult Intelligence Scale (WAIS), ability to withstand a tasty treat, highest degree obtained

About Variables: Variable

Variable:

- (Mathematical) placeholder for specific values
- E.g., `WAIS` is a variable representing scores on the WAIS
- You can refer to a variable without (yet) knowing its specific values

Data:

- Specific values of a variable
- Example of data: Jen's score on the variable `WAIS` is 138

Example

- Construct: Happiness
- Operational definition: “On a scale from 1 to 10, to what extent do you consider yourself a happy person?”
- Variable: `happiness`, taking values 1–10
- Data: respondent 382 answered 8

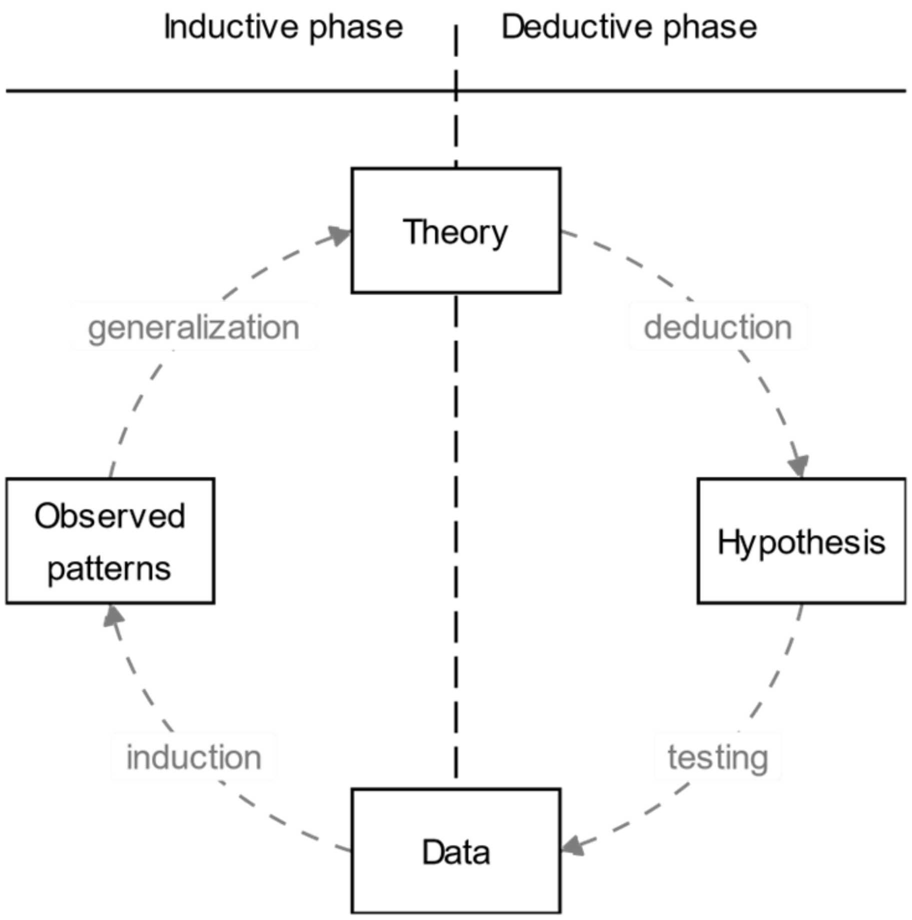
Break

See you in ~10 minutes!

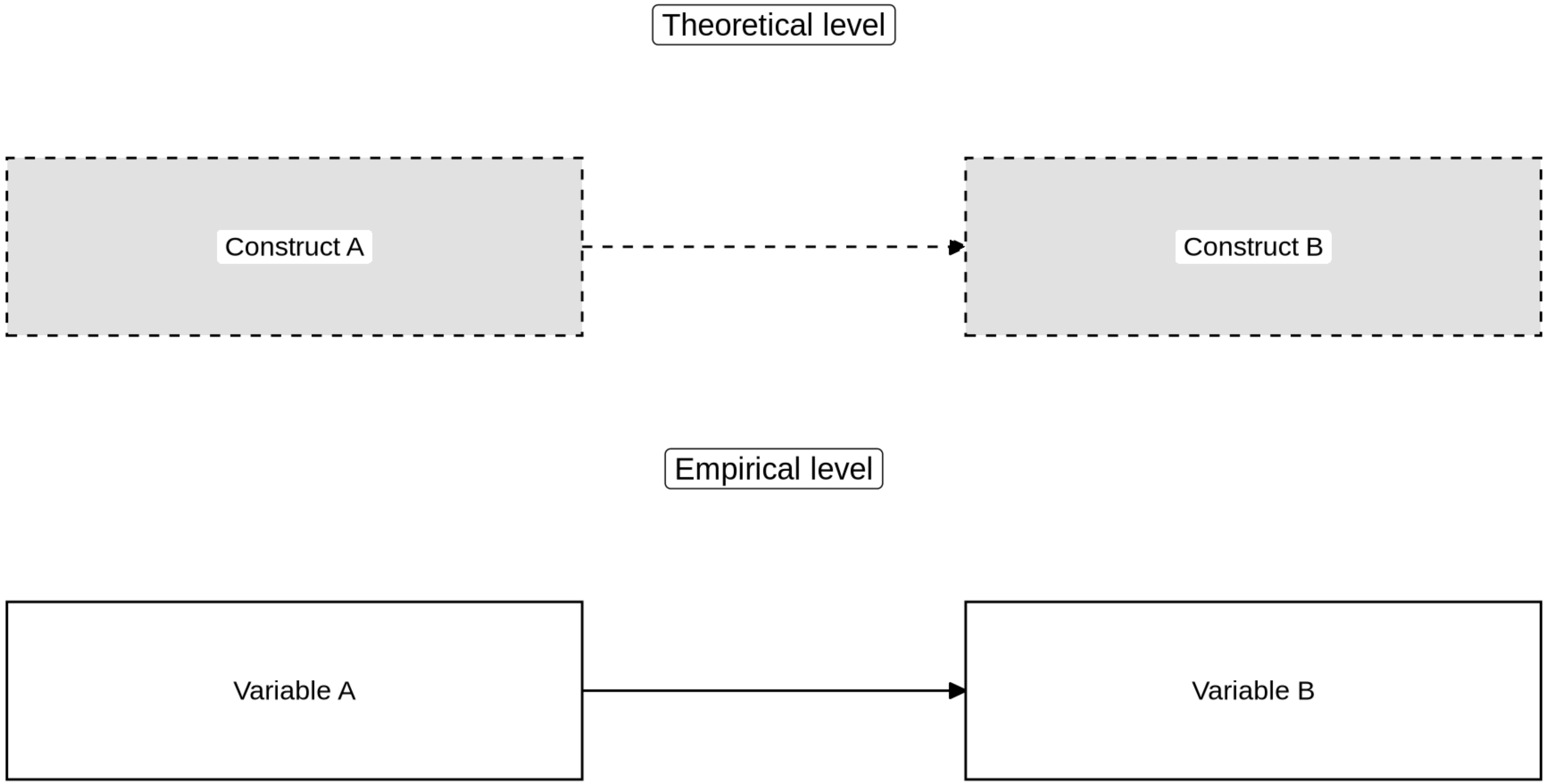
Role of Data in Scientific Research

Empirical cycle

De Groot (1961)



Relationship between Theory and Empiricism



Relationship between Theory and Empiricism

Theory:

Construct A influences construct B

- E.g., being exposed to war increases depression

Empirical test:

Variable A predicts variable B

- E.g., being deployed (yes/no) predicts scores on the SCL-90 depression questionnaire

Relationship between Theory and Empiricism

CBS reports that 87.4% of Dutch adults are “happy”. What does “happy” mean?

- CBS classifies a responses as:
  - 7–10: happy
  - 5–6: as neither happy nor unhappy
  - 1–4: unhappy

Note: Happiness is not “answered >6 on this particular question”.

This is just a (necessary) operationalization.

Theory: Having meaningful social relationships makes people happier.

Empirical hypothesis: Respondents who have frequent contact with family/friends report higher happiness scores.

Sampling Theory

Population → Random sample → Sample

Sampling theory

Population

Variable X

Population data → Often not available

Data-analysis (***descriptive statistics*** of big dataset):

- Parameters like μ, σ^2 and π
- *Indicated with Greek letters*

Sample

Variable X

Sample data

Data-analysis (***descriptive statistics*** of smaller dataset):

- Statistics like $\bar{x}$, s^2 and p
- *Indicated with Roman letters*

Inferential statistics

Draw conclusions about *population* parameters on basis of *sample* statistics .

Samples: Why?

- We want to make a claim about the population
- It is not possible to access the entire population
- We observe a part of the population: the *sample*
- Sample statistics are also our best guess about population parameters
  - Of course, sample statistics are not *equal to* population parameters
  - You will learn ways to express uncertainty about your best guess

Samples: Why?

- If the sample is *representative* of the population, your best guess will generalize better to the entire population and to new samples
- The best way to get a representative sample is *random sampling*
- Random sample:
  - Every population individual has the same probability of being included

Non-random samples

- Probability of inclusion unknown to the researcher
- Examples: convenience sampling, snowball sampling, cluster sampling
- Sampling bias
  - E.g., Individuals that are easy to access ave higher probability of being included

A lot of sampling theory is just *cooking*

- Suppose you want to know if your soup needs more salt.
  - You don't eat the whole soup, you *sample* it.
  - You stir the soup so to take a *random* sample.
  - If you keep tasting from the spoon while adding salt to the soup, the sample is *non-random*.

Measurement Level

NOIR measurement levels

Measurement level: What kind of information is contained in a variable?

Mnemonic: `n o i r` = black in French

Subsequent levels carry incremental information

The measurement level determines what statistics are appropriate. - Example: We cannot compute a mean for gender

NOIR measurement levels

- `N`ominal
  - Categories; values differ in name only (e.g., majors)
- `O`rdinal
  - Categories; additionally, values have meaningful order (e.g., SES groups, bachelor year 1, 2, 3)
- `I`nterval
  - Numeric; additionally, distance between values is meaningful
  - A step from 1 to 2 is equally large as a step from 2 to 3, or 5 to 6
- `R`atio
  - Numeric; additionally has a meaningful zero-point
  - Because of this, ratios between two values are also meaningful

Example

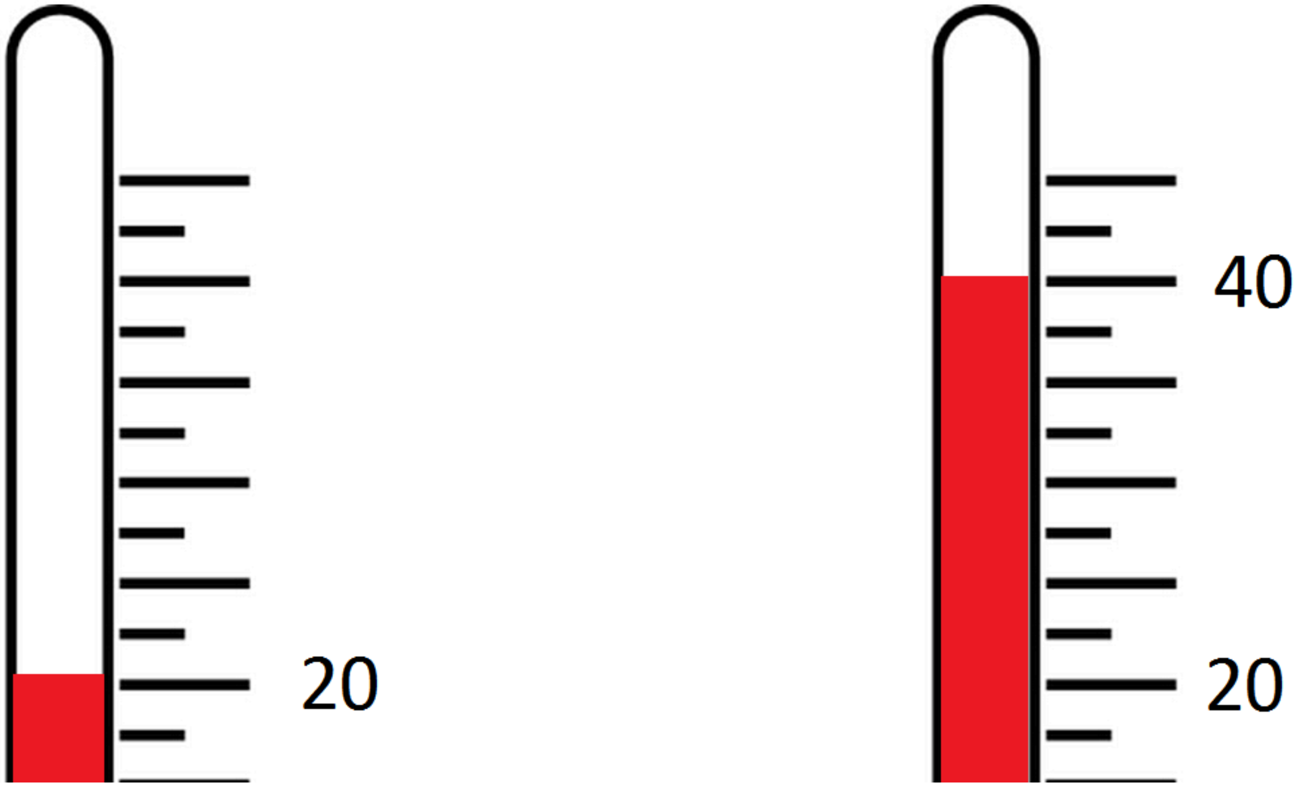
- Operational definition of happiness
    - “On a scale from 1 to 10, to what extent do you consider yourself a happy person?”

What measurement level is this?

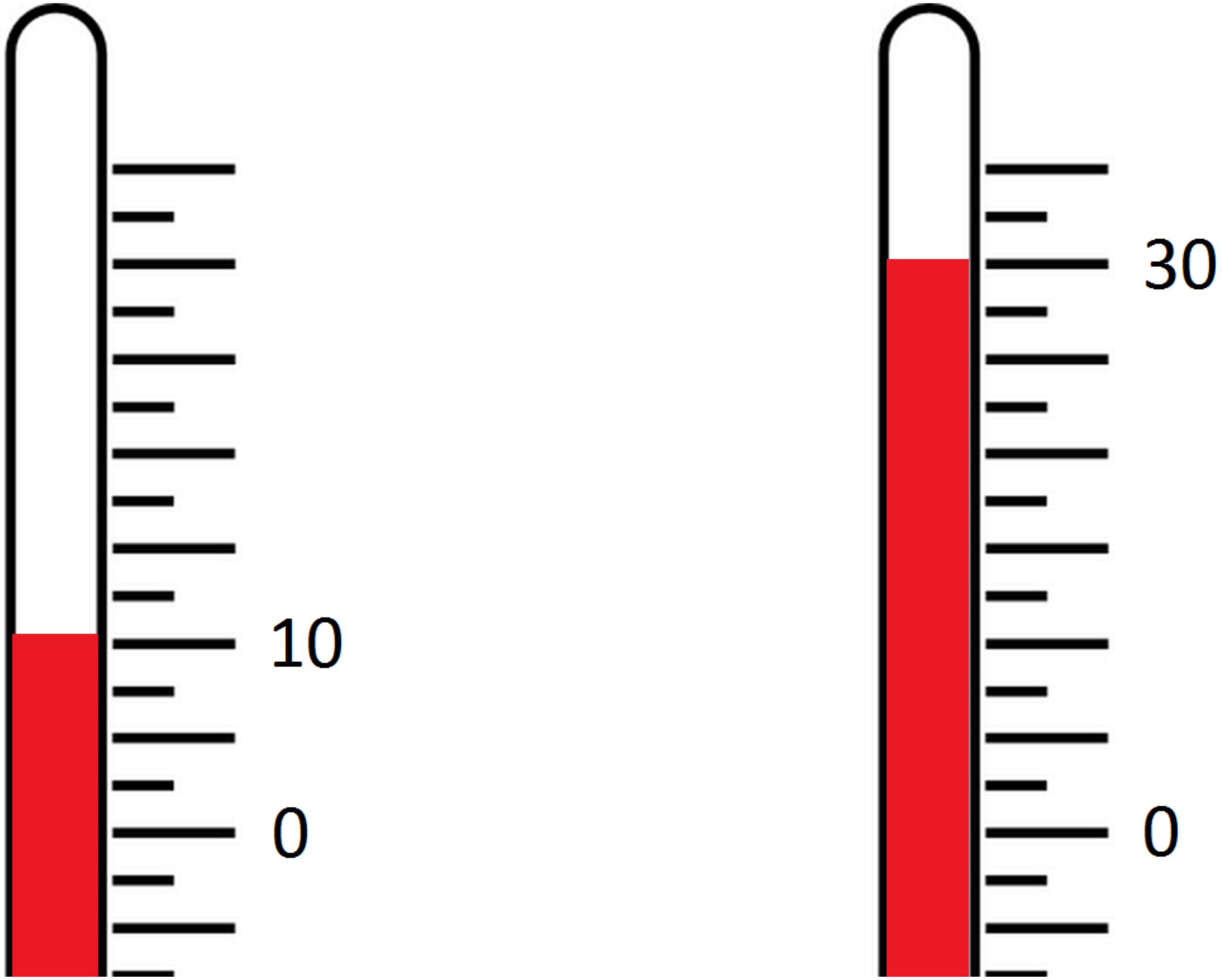
Ordinal because distances are meaningless. In practice often analyzed as interval (which is suboptimal).

Interval v Ratio

- Celsius and Fahrenheit are interval scales
- Kelvin is a ratio scale



Interval v Ratio



Measurement level matters

- Measurement level is a property of the construct and of the operational definition
  - Ideally, the measurement level of the construct and its variable are the same
  - E.g., sex assigned at birth: nominal
  - E.g., gender identification: ordinal or continuous
- Measurement level determines what statistics and analyses you can use

Other common distinctions

- Categorical variables: Nominal, Ordinal
- Continuous variables: Interval, Ratio
- Qualitative variable: Difference in kind, Nominal
- Quantitative variables: Differences in degree; ordinal, interval, ratio

Edge cases:

- Number of children (discrete ratio)
- Political orientation from liberal to conservative (ordinal, but which is higher/lower?)

Example

What measurement level is age?

- Milliseconds since the big bang?
- Unix time (the number of non-leap seconds that have elapsed since 00:00:00 UTC on 1 January 1970)
- Intervals: 0 - 5, 6 - 10, 11 - 15
- Months
- Generation: Generation X, Millennials, Gen Z, Gen Alpha

It depends on the way it's measured!

Descriptive Statistics

Descriptive Statistics

- Descriptive statistics summarize data across a sample
- You nearly always examine them to get a sense of your dataset
  - E.g., which major is most common among LAS students?
  - How old are my students, on average?
  - How much do the ages of my students vary?
  - What’s the gender distribution?
- Descriptive statistics *may* also be relevant to answer research questions
  - E.g.: When evaluating exam questions: Is the proportion of correct answers on this MC question greater than chance?

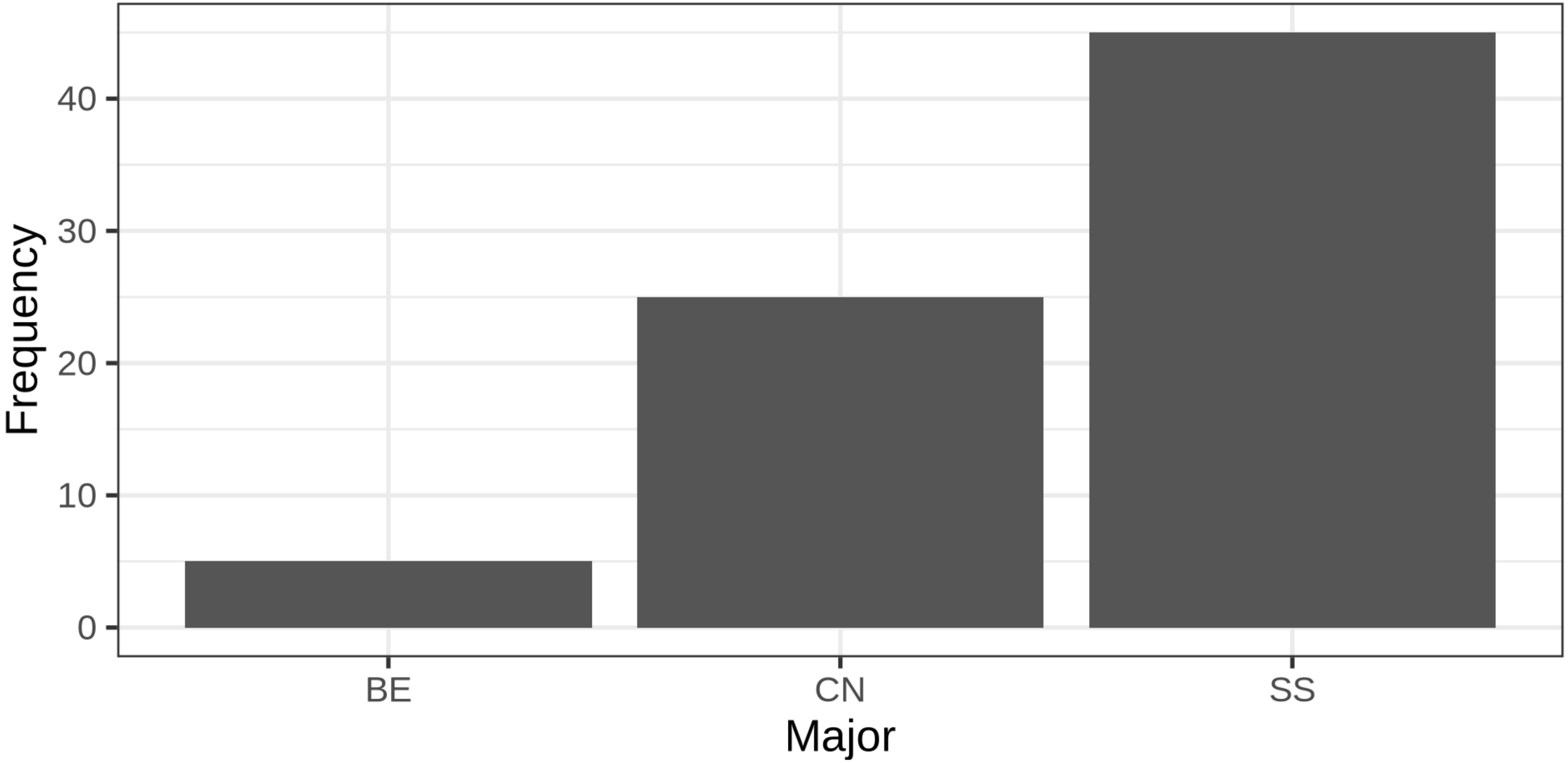
Distributions

TYPE OF VARIABLE	CENTRAL TENDENCY	GRAPH
Nominal	Frequency distribution	Bar chart
Ordinal	(Cumulative) frequency distribution	Bar chart
Interval/ratio	(Normal) probability distribution	Histogram, density plot

A Nominal Variable

MAJOR	FREQUENCY	PERCENT
BE	5	0.07
CN	25	0.33
SS	45	0.60

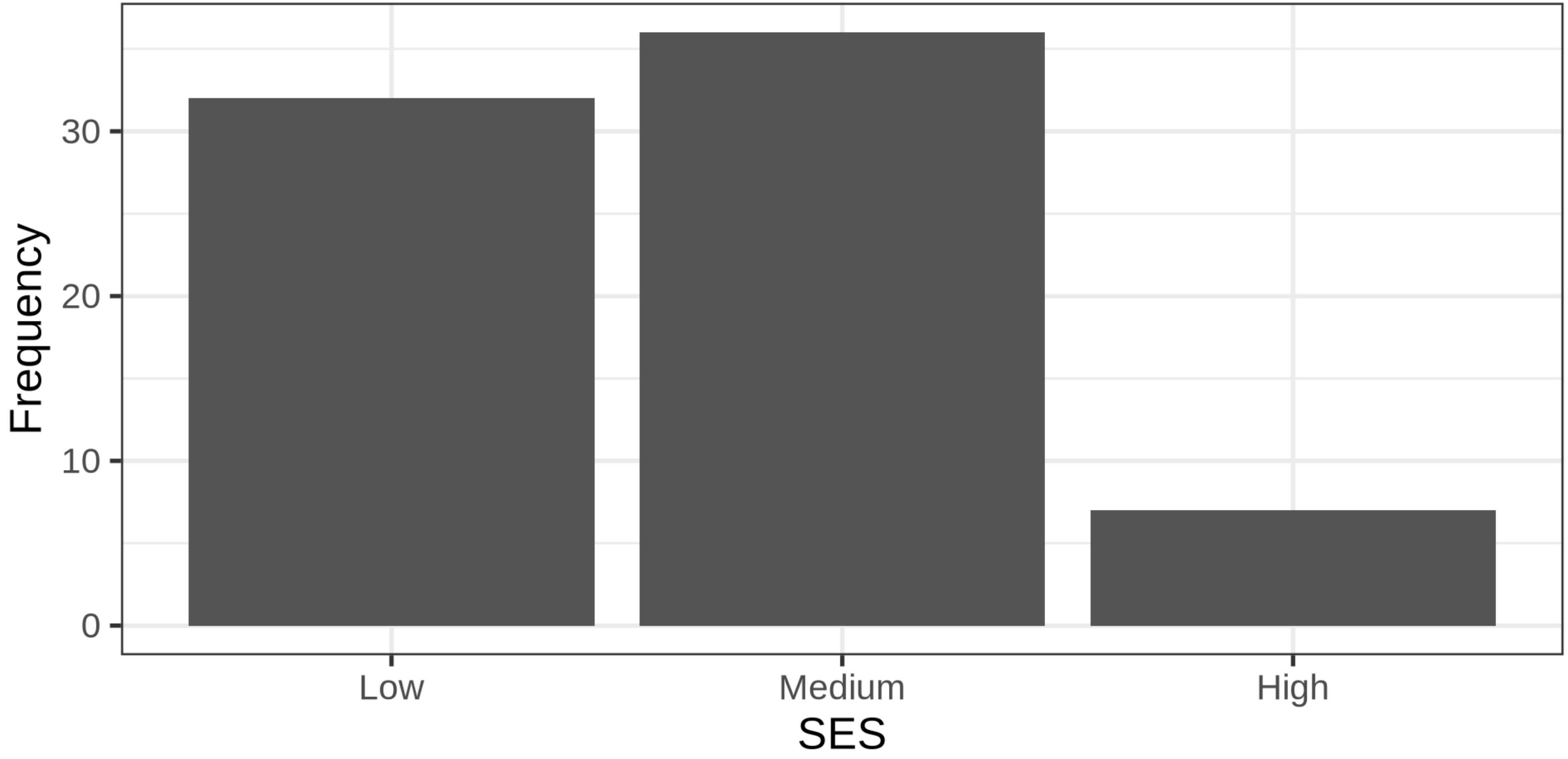
Bar Chart



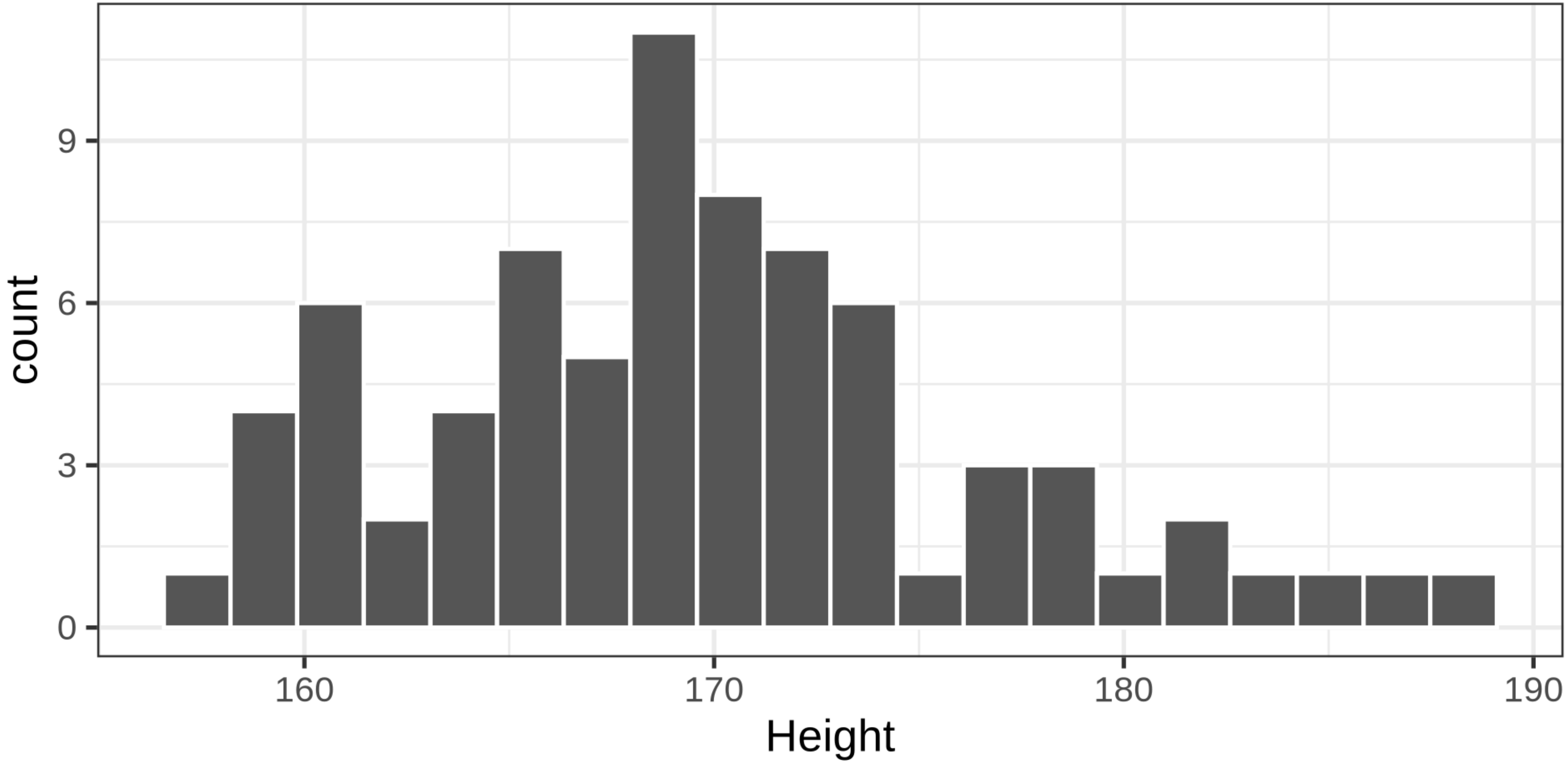
An Ordinal Variable

SES	FREQUENCY	PERCENT	CUMULATIVE
Low	32	0.43	0.43
Medium	36	0.48	0.91
High	7	0.09	1.00

Bar Chart



A Continuous Variable



Measures of Central Tendency

What is the average or most common response?

TYPE OF MEASURE	TYPE OF VARIABLE	DEFINITION
Mode	Nominal/Ordinal	Most common value
Median	Ordinal/Continuous	Middle value/50th percentile
Mean	Continuous	Average value

Measures of Dispersion

How much variability is there in responses?

TYPE OF MEASURE	TYPE OF VARIABLE	DEFINITION
Frequency table	Nominal	Count/percentage of responses
Range	Ordinal/continuous	Minimum to maximum value
Variance	Continuous	Mean squared distance of observations to the mean

Calculating the Variance

Variance:

$$S_X^2 = \frac{\sum_{i=1}^{n} (X_i - \bar{X})^2}{n-1}$$

Units of S^2 are squared; if you measured height in meter m, S^2_{height} is expressed in m^2

Standard deviation (SD): $S_X = \sqrt{S_X^2}$, so the unit is m.

For what variables is this meaningful?

Interval and ratio variables.

Median and Mode for Ordinal Variable

Median: middle value

Example 1: (n = unequal): 4, 5, 6, 7, 8, 9, 9 -> median is 7

Example 2: (n = equal): 4, 5, 6, 8, 9, 9 -> median 7 (mean of the two middle values)

Mode: Most frequent score; 9 in both examples

Mean, Median, Mode for Continuous Variable

The average Australian is a millionaire


COLUMN
IONICA SMEETS

Australiërs zijn gemiddeld miljonair, maar daar hebben de meeste Australiërs helemaal niks aan

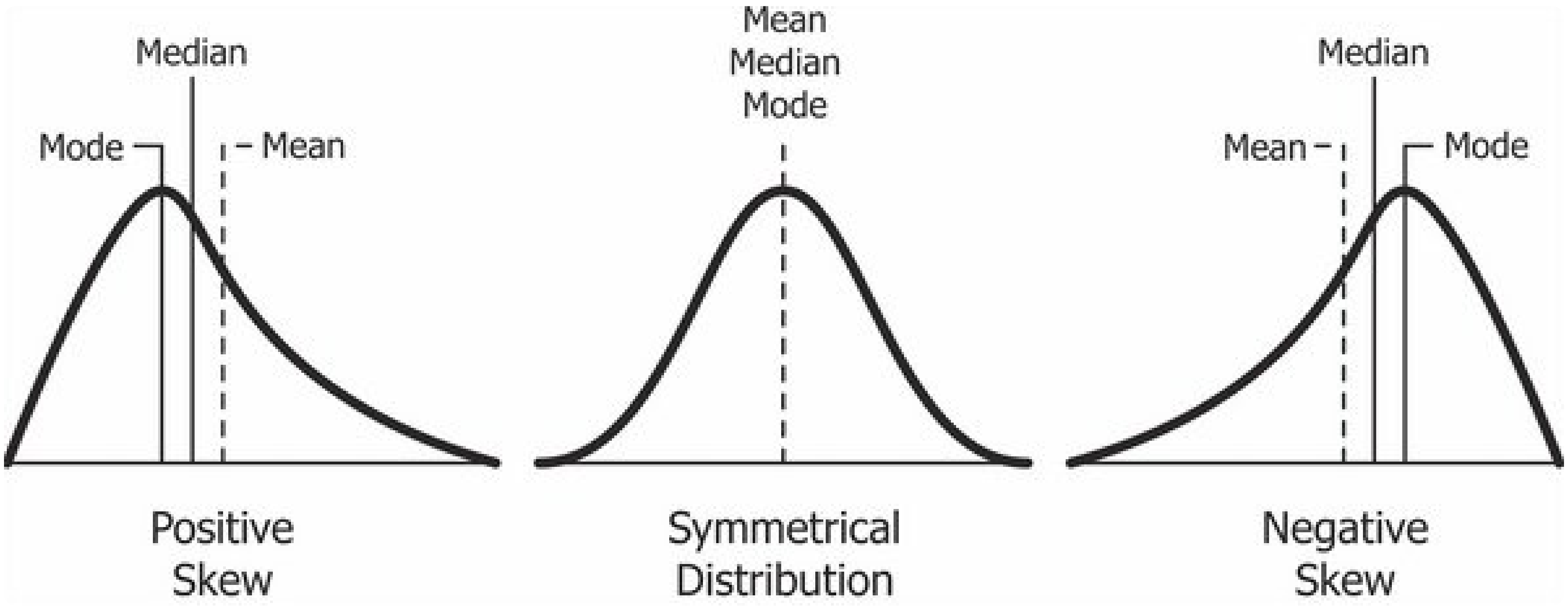
23 augustus 2019

G oed nieuws voor Australiërs deze zomer: ze zijn

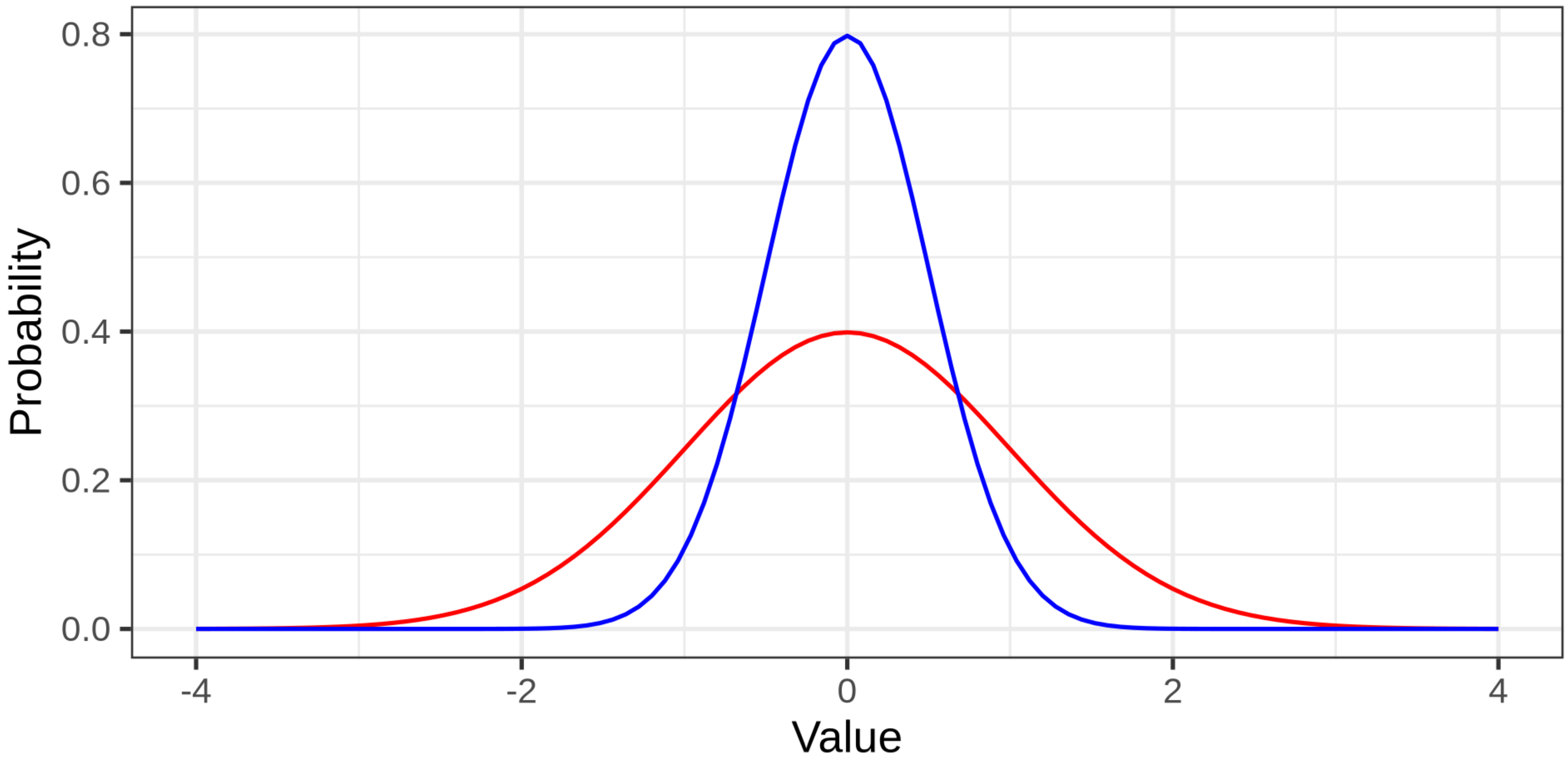
But... most Australians are not!

Source: https://www.volkskrant.nl/columns-opinie/australiers-zijn-gemiddeld-miljonair-maar-daar-hebben-de-meeste-australiers-helemaal-niks-aan~b8aa3fd3 without paywall

Skewed Distributions



Mean, variance



Bivariate Descriptives

Describing associations

	NOMINAL	ORDINAL	INTERVAL/RATIO
Nominal	Contingency Table	Contingency Table	Contingency table
Ordinal		Contingency Table Spearman's correlation	Biserial Correlation
Interval/Ratio			Pearson Correlation Coefficient Scatter plot

Contingency table

Sexe * Education Crosstabulation

			Education			
			Low	Middle	High	Total
Sexe	Female	Count	40	20	15	75
		% within Sexe	53.3%	26.7%	20.0%	100.0%
	Male	Count	10	20	50	80
		% within Sexe	12.5%	25.0%	62.5%	100.0%
Total		Count	50	40	65	155
		% within Sexe	32.3%	25.8%	41.9%	100.0%

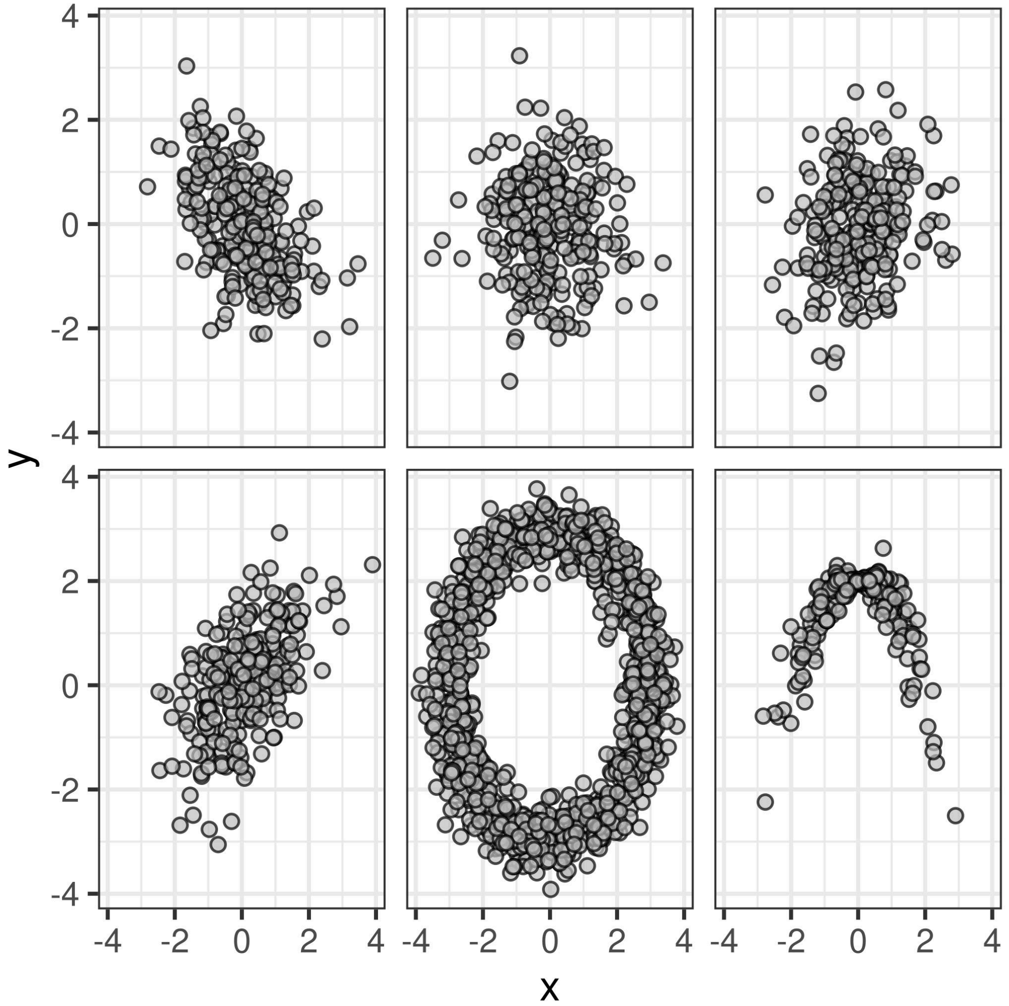
Is there an association between gender and education?

Correlation

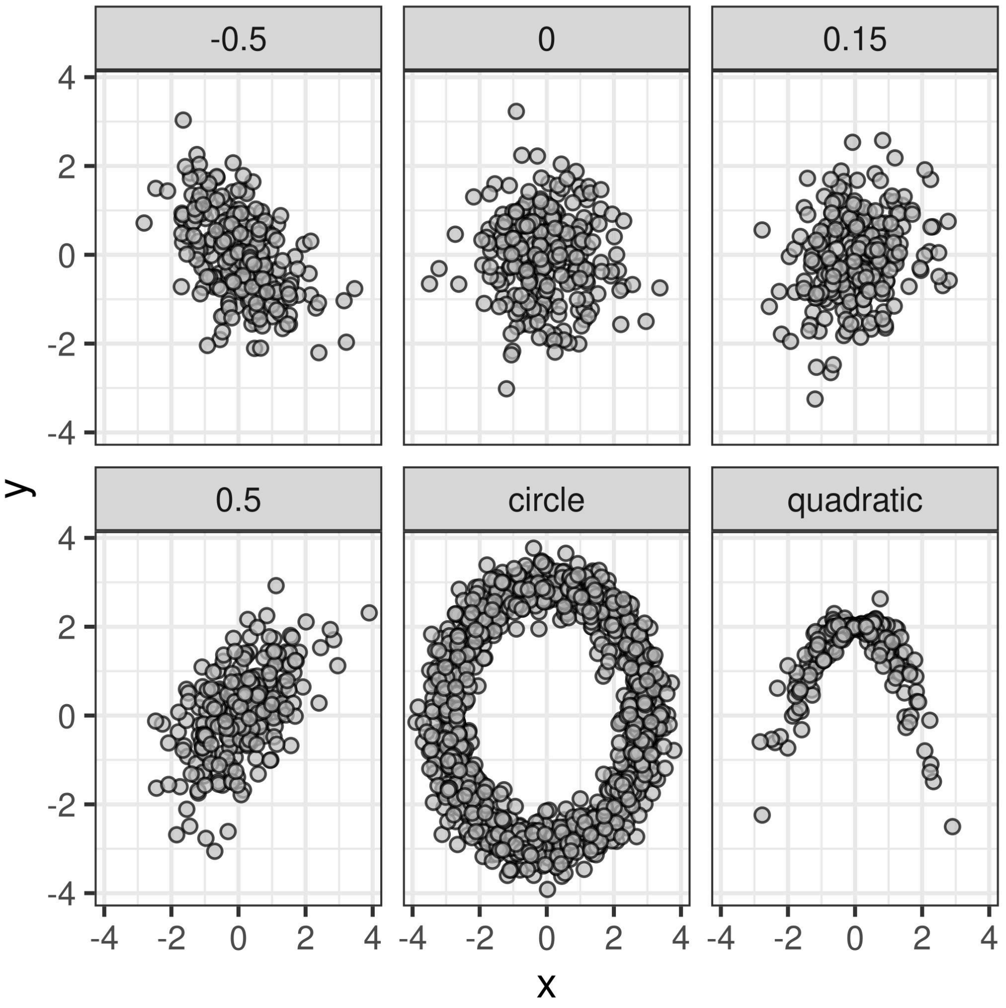
> **Correlation:** A standardized measure of the strength of linear association between two continuous variables.

- Standardized: ranges from $[-1, 1]$
- Sample correlation: r
- Population correlation: ρ
- $r = -1$: Perfect negative association
- $r = 0$: No association
- $r = 1$: Perfect positive association

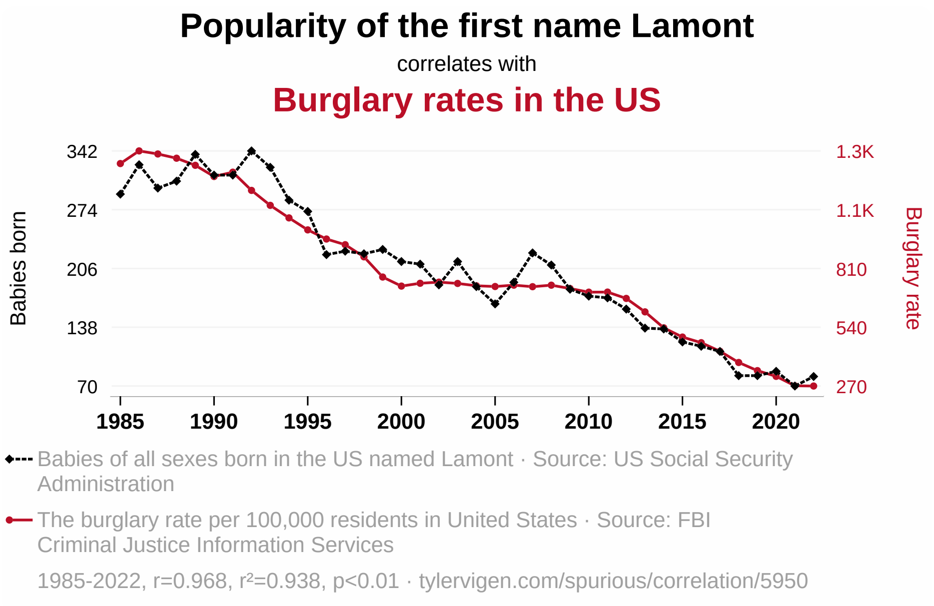
Correlations

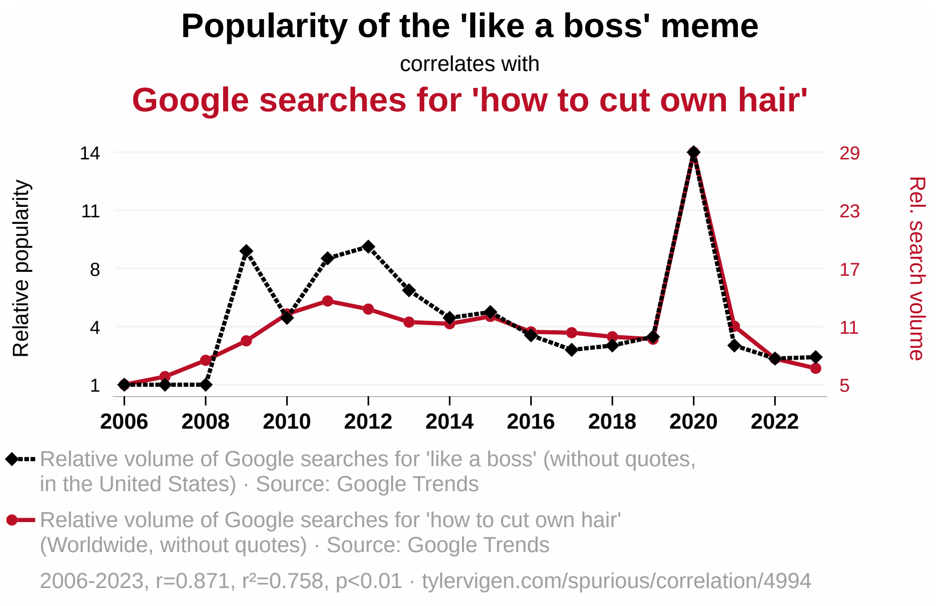


Correlations



Correlations





Correlation is not causation!

Source: left right

Recap

You now know:

- The difference between methods and statistics
- The difference between the population and a sample
- The different measurement levels
- A few univariate and bivariate descriptives statistics

See you next week!